

Sex-specific associations of steatotic liver disease with vascular health

To the Editor:

Given the importance of steatotic liver disease (SLD) in relation to cardiovascular disease (CVD) risks, we have followed with great interest the recent advances in the field including SLD nomenclature modifications to replace non-alcoholic fatty liver disease (NAFLD) with metabolic dysfunction-associated steatotic liver disease (MASLD).¹ In particular, we appreciate the inclusion of “metabolic” in the terminology and addition of cardiometabolic diagnoses within the diagnostic criteria given the need to improve our understanding of the disease and its sequelae. Awareness of NAFLD as a highly prevalent and potent cardiometabolic risk factor was only beginning to emerge within the cardiovascular field prior to publication of nomenclature changes by the *Journal of Hepatology*.² We anticipate this recognition will continue to grow for several reasons. First, a subsequent report in the *Journal* has demonstrated strong consistency between MASLD and NAFLD clinical profiles and outcomes.³ Second, the FDA recently approved resmetirom for metabolic dysfunction-associated steatohepatitis (MASH),⁴ and this agent appears to confer significant positive effects on atherogenic lipid profiles in addition to MASH.⁵ Third, obesity-related medications including GLP-1 receptor agonists have been shown to improve both hepatic and cardiovascular risk resulting in a broadening CVD prevention framework within therapies for traditionally “metabolic-related” conditions.⁶ Given these rapid advances in the field, we expect an increasing need for collaboration between hepatologists and cardiologists to address SLD and its epidemiological sequelae. In this context, we wish to raise considerations related to the cardiovascular impact of SLD that may be less immediately apparent. Our observations arise from having examined the effects of standardized diagnostic thresholds for SLD and hepatic fibrosis on vascular health in males and females.

We studied 5,119 adult participants aged 18 years and older of the National Health and Nutrition Examination Survey (NHANES) 2017–2018 visit, who completed vibration-controlled transient elastography measurements of controlled attenuation parameter (CAP) and liver stiffness measurements (LSM) serving as surrogates for steatosis and fibrosis, respectively (age 50 ± 18 , 50% female, CAP 264 ± 63 dB/m, LSM 6.0 ± 5.4 kPa). We plotted CAP and LSM in relation to systolic blood pressure (SBP) with 4-knot natural splines and used American Association for the Study of Liver Disease (AASLD) guideline thresholds⁷ to generate continuous piecewise linear regressions, comparing the segmental slopes between sexes using interaction terms. Significant sex differences were seen

overall, and particularly for CAP (Fig. 1). Notably, with a diagnostic cut-off point of 288 dB/m, there was a significantly steeper association between increases in CAP and increases in SBP in females prior to the cut-off and no significant difference between females and males after the cut-off (beta [SE] before cut-off: female 0.096 [0.011], male 0.033 [0.010], p -difference < 0.001 ; after cut-off: female 0.049 [0.026], male 0.067 [0.020], p -difference = 0.045). These pre-cut-off sex differences were driven by findings in the younger half of the population (age ≤ 50 years) showing significant differences in slope between females and males (female 0.073 [0.009], male 0.036 [0.010], p -difference = 0.006), and the older half (age > 50) having no significant sex difference (female -0.001 [0.017], male -0.022 [0.017], p -difference = 0.38). For LSM, slopes appeared similar between females and males, both within the natural splines and the piecewise linear models, with the most significant increases seen before the first cut-off (< 8 kPa: females 2.29 [0.32], males 2.03 [0.29], p -difference = 0.55; 8 to < 12 kPa: females 1.00 [0.86], males 0.40 [0.69], p -difference = 0.56; ≥ 12 kPa: females 0.11 [0.72], males -0.13 [0.54], p -difference = 0.12), and no differences seen in any of the segmental slopes after age stratification (p -difference > 0.05 for all).

Epidemiologic data from population-based studies have consistently demonstrated differences between females and males in vascular aging trajectories,⁸ and more recent studies have indicated that the vascular response to similar doses of chronic metabolic stress may be more pronounced in females than males over the life course.^{9,10} Thus, interactions between sex, SLD, and vascular aging trajectories warrant further attention, especially given their potential impact on the efficacy of treatments being developed across the cardiometabolic space. Notwithstanding the diversity of approaches to diagnosing both SLD and hepatic fibrosis, subgroup-specific risks, and possible non-linear associations between CAP and true hepatic steatosis, our observations to date can serve to highlight three key points: (1) the vascular effects of hepatic steatosis (even more so than hepatic fibrosis) differ by sex with an apparently earlier onset and steeper trajectory of vascular risk accumulation in females than males; (2) these differences are driven by findings that are especially pronounced in younger-aged adults, including persons who have yet to carry an overtly recognized diagnosis but who may well be within a phase of transition from metabolic health to metabolic disease; and, (3) the finding of sex differences in sensitivity to metabolic exposures implies there may well be sex differences in response to metabolic risk modulating therapies, such that ongoing efforts to develop interventions for SLD should

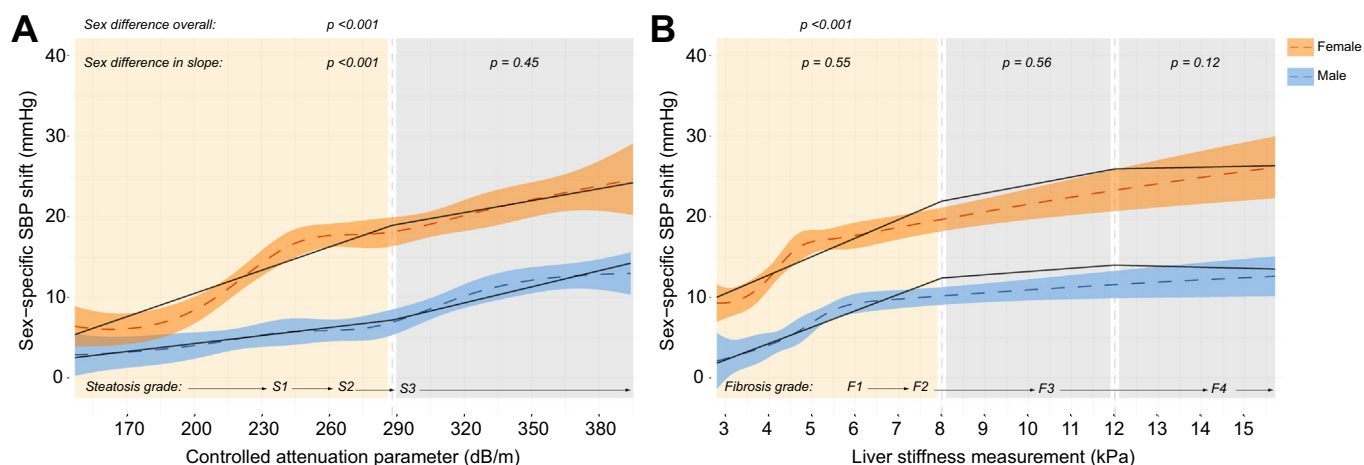


Fig. 1. Hepatic steatosis and hepatic fibrosis in relation to vascular health, by sex. Controlled attenuation parameter (A) and liver stiffness measurement (B) serve as surrogates of hepatic steatosis and hepatic fibrosis, respectively. *p* values are shown for the sex difference in overall trend and in the segmented slope by piecewise linear regression of rise in SBP, serving as a measure of vascular health, across categorical liver measures. In these analyses, we observe that females demonstrate greater vascular sensitivity to accumulation of either hepatic steatosis (left panel, A) or hepatic fibrosis (right panel, B) overall. Moreover, females also demonstrate earlier-onset vascular sensitivity to accumulation of hepatic steatosis, in particular, as reflected by a significantly steeper initial rise in SBP (i.e. difference in slope) from the sex-specific normal healthy (i.e. physiologic) thresholds (110 mmHg for females, 120 mmHg for males) previously established from multi-cohort epidemiologic reference data.⁸ SBP, systolic blood pressure.

consider sex-differential risk patterns as part of efforts to optimize treatment outcomes for all.

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Conflict of interest

Alan Kwan reports NIH grant funding (NIH-UL1TR001881). Minhao Wang has no conflicts to report. Hirsh Trivedi reports consulting fees from Novo Nordisk and Cedars-Sinai support for meeting attendance. Hongwei Ji has no conflicts to report. Susan Cheng reports NIH grant funding (NIH- U54-AG065141).

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Alan Kwan was involved in conceptualization, methodology, formal analysis, investigation, writing- original draft, writing-review & editing, visualization, and funding acquisition. Minhao Wang is involved in methodology, formal analysis,

writing- original draft, writing-review & editing, and visualization. Hirsh Trivedi is involved in conceptualization, investigation, and writing-review & editing. Hongwei Ji is involved in conceptualization, writing-review & editing, and supervision. Susan Cheng is involved in conceptualization, methodology, formal analysis, investigation, writing- original draft, writing-review & editing, visualization, supervision, and funding acquisition. Alan Kwan and Susan Cheng had direct access to the data and verify the underlying data reported.

Data availability statement

The NHANES data were acquired from and are available to the scientific community through <https://www.cdc.gov/nchs/nhanes/index.htm>. Data was accessed on 4/1/2024 through R Package [nhanesA](#).

Ethical approval of studies and informed consent

The NHANES protocol for 2017-2018 was approved under Protocol #2018-01 and Continuation of Protocol #2011-17.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.05.003>.

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